

Background	Results									
What it is: Dyslexia is a language-based learning disability linked to phonological processing difficulties and neurobiological brain differences (e.g., increased myelination in the left perisylvian cortex; Munzer et al., 2020). Why it matters: It can cause unexpected reading challenges, especially with reading fluency and word recall (Yale Center for Dyslexia & Creativity). How common / what we don’t know: Dyslexia affects about 20% of the population, and many aspects are still not fully understood (Yale Center for Dyslexia & Creativity). Most research is focused on K-12 student. Research gap: Most dyslexia research focuses on K–12 students; while some assistive technologies (e.g., text-to-speech/audio tools) show promise, there is little research on adults—and virtually none on workplace text-to-speech use.	Independent means <i>t</i> -test Comparing Groups Dyslexia (6) and Non-Dyslexia (149)									
	Outcome	No Dyslexia Mean	No Dyslexia (SD)	Yes Dyslexia Mean	Yes Dyslexia (SD)	<i>t</i>	<i>df</i>	<i>p</i>		
	Basic Needs Satisfaction	5.04	0.92	4.69	1.09	0.93	153	.36		
	Basic Needs Satisfaction Autonomy	5.32	1.09	5.22	0.78	0.21	153	.83		
	Basic Needs Satisfaction Competence	5.59	0.98	4.67	1.38	2.22	153	.03		
	Workplace Relatedness	4.89	1.33	4.54	2.08	0.62	153	.54		
	Workplace Competence	5.78	0.92	5.58	0.52	0.89	6.35	.41		
	Workplace Competence Frustration	2.24	1.07	2.92	1.24	-1.52	153	.13		
	Work Climate	5.19	1.28	4.85	0.48	1.57	8.23	.16		
	(2x2 standard factorial ANOVA) Dyslexia Status (Descriptive statistics) speed					(2x2 standard factorial ANOVA) Dyslexia Status (descriptive statistics) Grade				
Dyslexia Status	Condition	N	Mean	SD	Grade	Condition	N	Mean	SD	
No Dyslexia	Non-TTS	74	44.57	14.16	No Dyslexia	Non-TTS	74	6.08	2.08	
No Dyslexia	TTS	75	52.89	14.18	No Dyslexia	TTS	75	6.68	6.08	
Yes Deslexia	Non-TTS	2	92.06	68.62	Yes Deslexia	Non-TTS	2	8.00	1.41	
Yes Dyslexia	TTS	4	51.52	14.43	Yes Dyslexia	TTS	4	5.50	1.91	
Total	Non-TTS	76	45.82	17.79	Total	Non-TTS	76	6.13	2.09	
Total	TTS	79	52.82	14.10	Total	TTS	79	6.62	5.94	
Grand Total		155	49.39	16.34	Grand Total		155	6.38	4.48	
Effect (Descriptive statistics) (2x2 standard factorial ANOVA) speed					Effect (descriptive statistics) (2x2 standard factorial ANOVA) Grade					
Effect	<i>df</i>	<i>F</i>	<i>p</i>	<i>np</i> ²	Grade	<i>df</i>	<i>F</i>	<i>p</i>	<i>np</i> ²	
Dyslexia Status	1, 151	11.87	< .001	.07	Dyslexia Status	1, 151	0.04	.85	.00	
Condition (Non-TTS vs TTS)	1, 151	5.79	.02	.04	Condition (Non-TTS vs TTS)	1, 151	0.23	.63	.00	
Deslexia × Condition	1, 151	13.32	< .001	.08	Deslexia × Condition	1, 151	0.61	.44	.00	
Limitations										
Small dyslexia sample: Lower power; less stable estimates.										
Small subgroup: Interaction (esp. speed) may be outlier-sensitive.										
Speed assumptions: Unequal variances weaken ANOVA confidence.										
Generalizability: Replicate with a larger, balanced sample.										